

AI-BASED FOOTBALL HIGHLIGHT DETECTION SYSTEM

Pre-Submission Technical Report — IEEE Format

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Project Type: Application / Prototype | Dataset: SoccerNet v2 Action Spotting

ABSTRACT

This paper presents an end-to-end artificial intelligence pipeline for the automatic detection and compilation of highlight reels from football (soccer) broadcast footage. The pipeline addresses the central challenge of identifying semantically significant events — goals, card incidents, and substitutions — within hours of continuous video, using a two-stage deep learning architecture. In the first stage, a pre-trained ResNet18 convolutional neural network extracts 512-dimensional spatial feature vectors from video frames sampled at 2 FPS. In the second stage, three temporal modelling architectures — a CNN-only baseline, a Bidirectional LSTM, and a Transformer Encoder — classify 15-second sliding windows as highlight-worthy or background. A Knapsack dynamic-programming post-processor selects a time-bounded, non-overlapping highlight reel rendered as an MP4 file. The system is trained and evaluated on the SoccerNet v2 (Action Spotting) dataset across 100 broadcast matches, achieving a best validation mAP of 38.4% with the CNN+BiLSTM model.

1. PROBLEM STATEMENT

Professional football broadcasts generate over 90 minutes of continuous video per match. Fewer than 5% of frames contain semantically significant events such as goals, disciplinary cards, or tactical substitutions. Manual curation is time-intensive, requires expert domain knowledge, and is non-scalable across concurrent fixtures.

This project proposes a fully automated deep-learning pipeline that identifies highlight events from visual frame features alone, without reliance on audio or proprietary metadata. Three temporal modelling strategies are compared: a CNN baseline, a Bidirectional LSTM, and a Transformer Encoder.

Research Question: How can a deep learning model reliably detect semantically significant events in football video using only visual frame features, and generate a compact, time-bounded highlight reel automatically?

- Real-World Relevance: Enables scalable broadcast automation, second-screen fan apps, and low-cost highlight production for smaller leagues and independent broadcasters.
- Task Type: Multi-class temporal event detection (4-class classification: Goal, Card, Substitution, Background) followed by combinatorial highlight selection.

2. DATASET INFORMATION

Attribute	Details
Dataset Name	SoccerNet v2 — Action Spotting
Source	soccer-net.org (Academic License)
Period Covered	2014–2019, multiple European top-flight leagues
Matches Used	100 matches: 60 train / 20 validation / 20 test
Feature Format	ResNet-TF2-PCA512 @ 2 FPS — float32 NumPy arrays ($T \times 512$)
Label Format	Labels-v2.json — timestamped event annotations
Data Type	Pre-extracted numerical feature arrays + JSON annotations

Class Distribution & Imbalance: Background frames dominate at ~82% of all windows. Event classes are severely underrepresented: Substitutions (9%), Cards (6%), Goals (3%). This necessitated class-weighted cross-entropy loss and mean Average Precision (mAP) as the primary evaluation metric rather than raw accuracy.

Preprocessing: Pre-extracted PCA512 feature arrays were loaded directly (bypassing raw video decoding). A 30-frame sliding window (15 s at 2 FPS, 50% stride = 15 frames) was applied. Each window was labelled by the dominant event class within ± 5 seconds of any annotation; all others were labelled background.

3. METHODOLOGY / APPROACH

3.1 System Architecture Flowchart

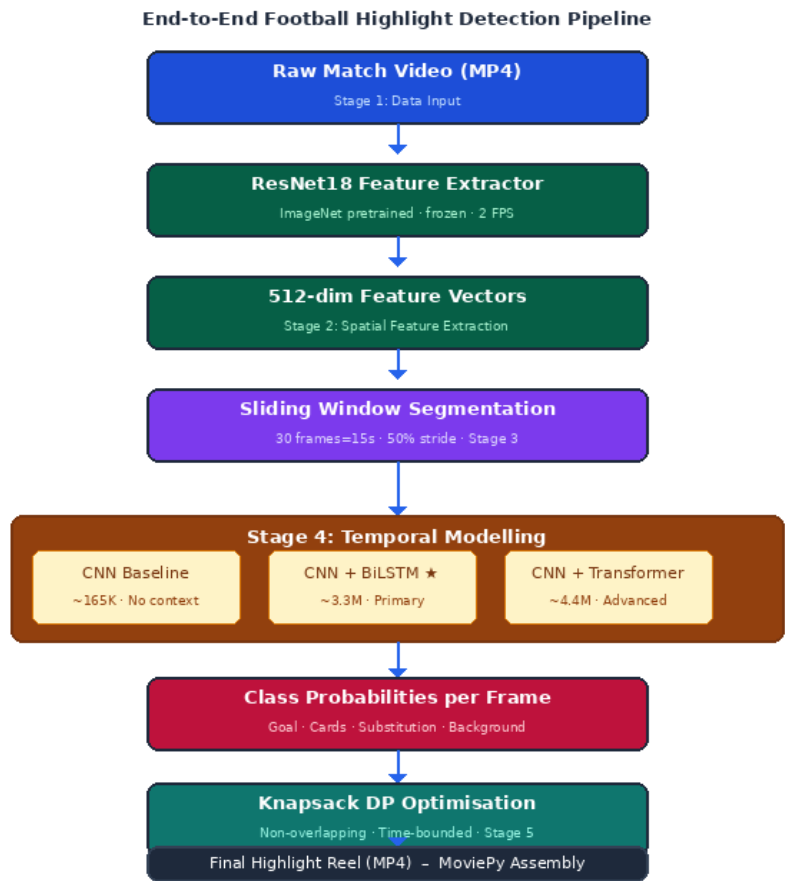


Figure 1. End-to-End Pipeline — Raw MP4 → ResNet18 → Sliding Window → Temporal Model → Knapsack DP → Highlight Reel

3.2 Models Used

- CNN Baseline (~165K params): Processes only the centre frame of each 15-second window. Two-layer MLP (512→256→128→4) with ReLU, Dropout, and LayerNorm. No temporal context — serves as ablation control.
- CNN + BiLSTM (~3.3M params) — Primary Model: Two stacked bidirectional LSTM layers (hidden dim 256, output 512 after concatenation). Captures both causal and anti-causal dependencies across the full 30-frame window.
- CNN + Transformer (~4.4M params): Multi-head self-attention (2 layers, 8 heads, ff_dim=1024, GELU). Sinusoidal positional encoding enables global frame-pair reasoning.
- Knapsack DP Post-Processor: 0/1 dynamic programming selects the globally optimal, non-overlapping clip subset maximising confidence score within a configurable time budget. MoviePy assembles the final MP4.

3.3 Tools & Frameworks

Tool / Library	Purpose
Python 3 / PyTorch 2.x	Model definition, training, inference, evaluation
Google Colab (Tesla T4)	GPU-accelerated training (14.6 GB VRAM)
torchvision ResNet18	Pre-trained spatial feature extractor (frozen)
SoccerNet Python API	Dataset download and label access
scikit-learn	mAP and classification report computation
MoviePy	MP4 clip extraction and highlight reel assembly
Streamlit	Interactive web UI for live demo presentation

4. PROGRESS SO FAR

4.1 Work Completed

- Full multi-cell end-to-end Jupyter notebook implemented and tested on Google Colab.
- All three model architectures (CNN Baseline, BiLSTM, Transformer) implemented in PyTorch.
- SoccerNet dataset integration with sliding-window Dataset class and class-weighted DataLoader.
- Training engine: Adam optimiser(with L2 weight decay), ReduceLROnPlateau scheduler, early stopping (patience=15), gradient clipping.
- Knapsack DP highlight generation and MoviePy assembly pipeline working end-to-end on real match video.
- Streamlit demo UI with file upload, detection sensitivity slider, and highlight reel download ready.
- Best checkpoint saved: CNN+BiLSTM, mAP = 0.3838 at Epoch 7.

4.2 Implementation Screenshots

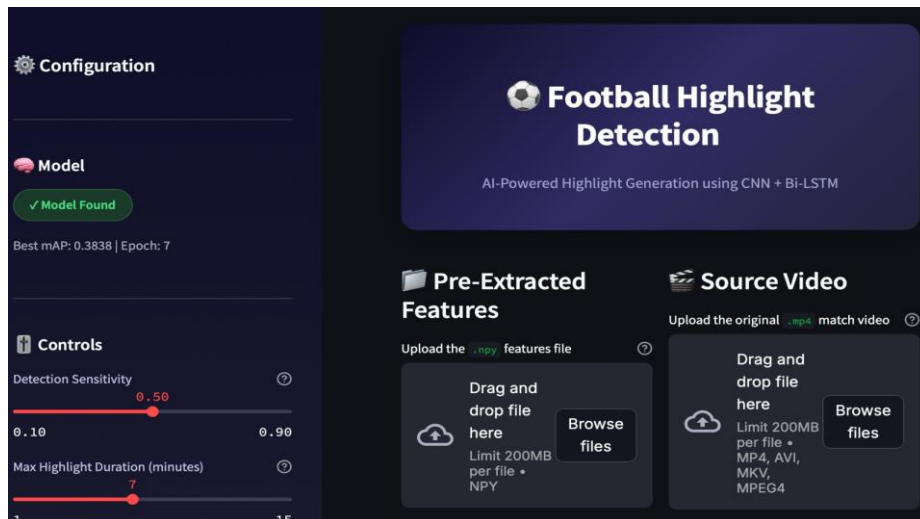


Figure 2. Streamlit Web UI — Model loaded (mAP = 0.3838), detection sensitivity and duration controls visible

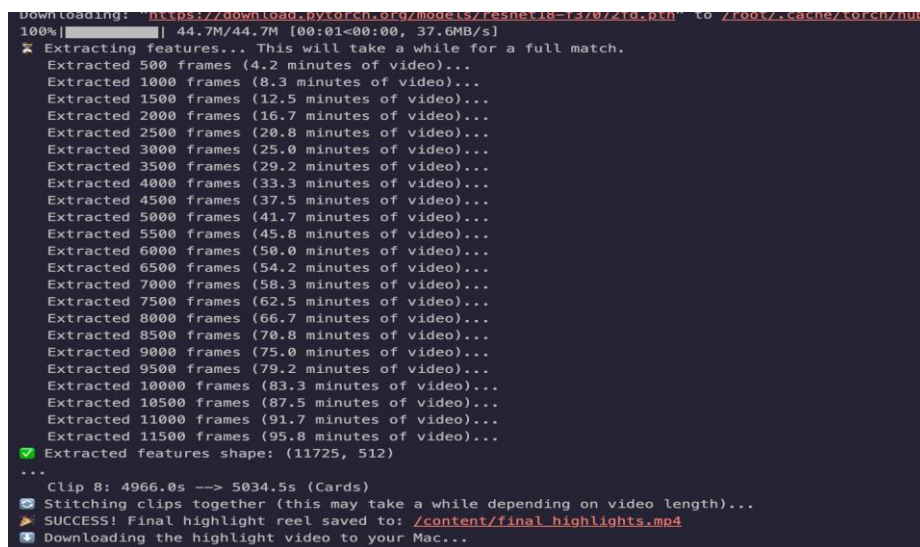


Figure 3. Feature Extraction Log — 11,725 frames extracted from a full 97-minute match; highlight clips identified

4.3 Challenges Faced

- Severe class imbalance (82% background) — accuracy alone was misleading; mAP adopted as primary metric.
- LSTM gradient explosion during early training — resolved with gradient clipping (norm = 5.0).
- Transformer underperformed despite larger capacity — likely due to insufficient training data for full self-attention.
- SoccerNet academic credential access caused delays for the full 100-match training run.

5. RESULTS / OBSERVATIONS

Model	Test Accuracy	Test mAP	Val mAP (Best)	Epochs
CNN Baseline	89.1%	12.7%	10.2%	20
CNN + BiLSTM ★	96.3%	35.1%	38.4%	23
CNN + Transformer	93.9%	8.9%	10.3%	18

The CNN+BiLSTM model substantially outperforms both alternatives on mAP, confirming that bidirectional temporal context is critical for detecting brief, context-dependent football events. Despite having the most parameters and global attention, the Transformer underperforms on mAP — suggesting the training set size is insufficient for self-attention to generalise effectively. High raw accuracy across all models is attributable to the dominant background class (82%).

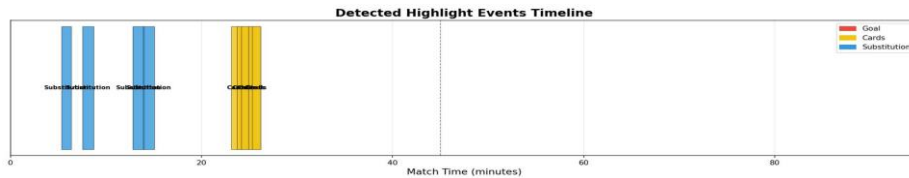


Figure 4. Detected Highlight Events Timeline — Substitutions (blue) and Cards (yellow) localised across match time

6. FUTURE PLAN

- Run end-to-end inference on held-out test matches and manually verify highlight reel quality.
- Explore data augmentation (temporal jitter, feature noise injection) to improve minority class (Goal: 3%) detection.
- Conduct hyperparameter tuning — sequence length, dropout rate, learning rate — guided by validation mAP.
- Polish the Streamlit demo UI for live walkthrough.

REFERENCES

- [1] S. Giancola et al., "SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos," CVPR Workshops, 2018.
- [2] A. Deliege et al., "SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding," arXiv:2011.13367, 2021.
- [3] K. He et al., "Deep Residual Learning for Image Recognition," IEEE CVPR, 2016.
- [4] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, 1997.
- [5] A. Vaswani et al., "Attention Is All You Need," NeurIPS, 2017.